Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **54**

Instructor Name: **Bodanske, Daniel**

Question 1 of 54

For each value of w, determine whether it is a solution to $7w - 3 = 32$.

w	Is it a solution?	
	Yes	**No**
-7	○	○
10	○	○
5	○	○
-2	○	○

Question 2 of 54

Solve for y.

$$-2 + 2y = -8$$

Simplify your answer as much as possible.

$y =$ _____

Question 3 of 54

Solve for u.

$$-5u + 6 = -12u + 62$$

Simplify your answer as much as possible.

Question 4 of 54

Solve for x.

$$3x + 36 = 6(x + 2)$$

Simplify your answer as much as possible.

$x =$

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Question 5 of 54

Jenny bought some books. She spent a total of $\$71$ after a discount of $\$5$ off of her total purchase. Each book cost $\$4$. How many books did she buy?

(a) Write an equation that could be used to answer the question above. First, choose the appropriate form. Then, fill in the blanks with the numbers 71, 5, and 4. Let b represent the number of books.

○ ___b + ___ = ___

○ ___b − ___ = ___

(b) Solve the equation in part (a) to find the number of books.

$b =$ _____

Question 6 of 54

Solve for x.

$$5xy = W$$

Question 7 of 54

Solve for n.

$$8m + 10n = -4$$

$n =$ ☐

Question 8 of 54

Solve the following proportion for y.

$$\frac{11}{5} = \frac{9}{y}$$

Round your answer to the nearest tenth.

$y =$ _____

Question 9 of 54

Solve for w.

$$\frac{w+2}{7} = \frac{5}{4}$$

Simplify your answer as much as possible.

$w =$ ☐

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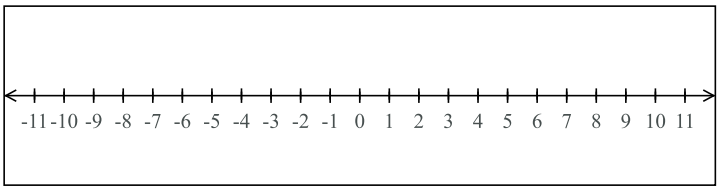
Question 10 of 54

Graph the inequality below on the number line.

$$a \leq -8$$



Question 11 of 54

Solve the inequality for v.

$$\frac{v}{-5} < -2$$

Simplify your answer as much as possible.

Question 12 of 54

Solve the inequality for y.

$$42 > 2y + 10$$

Simplify your answer as much as possible.

Question 13 of 54

Find the y-intercept and the slope of the line.

$$8x - 4y = 3$$

Write your answers in simplest form.

Question 14 of 54

Convert $77°F$ to degrees Celsius.
If necessary, round your answer to the nearest tenth of a degree.

Here are the formulas.

$$C = \frac{5}{9}(F - 32)$$

$$F = \frac{9}{5}C + 32$$

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Question 15 of 54

For each value of u, determine whether it is a solution to $28 = 1 - 9u$.

u	Is it a solution?	
	Yes	**No**
-7	○	○
9	○	○
-5	○	○
10	○	○

Question 16 of 54

Solve for y.

$$-24 = 5y + 6$$

Simplify your answer as much as possible.

$y =$ _____

Question 17 of 54

Solve for u.

$$8 = \frac{u}{6.25} + 3$$

$u =$ ☐

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Question 18 of 54

The equation $\frac{y}{3} - 12 = -7$ is solved in several steps below.

For each step, choose the reason that best justifies it.

Step	**Reason**
$\frac{y}{3} - 12 = -7$	Given equation
$\frac{y}{3} - 12 + 12 = -7 + 12$	-Addition Property of Equality -Subtraction Property of Equality -Multiplication Property of Equality -Division Property of Equality -Simplifying -Distributive Property
$\frac{y}{3} = 5$	-Addition Property of Equality -Subtraction Property of Equality -Multiplication Property of Equality -Division Property of Equality -Simplifying -Distributive Property
$3 \cdot \frac{y}{3} = 5 \cdot 3$	-Addition Property of Equality -Subtraction Property of Equality -Multiplication Property of Equality -Division Property of Equality -Simplifying -Distributive Property
$y = 15$	-Addition Property of Equality -Subtraction Property of Equality -Multiplication Property of Equality -Division Property of Equality -Simplifying -Distributive Property

Question 19 of 54

Solve for y.

$$11 + 15y = -7 + 13y$$

Simplify your answer as much as possible.

Question 20 of 54

Solve for x.

$$7(x + 6) = 4x + 36$$

Simplify your answer as much as possible.

$x =$

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Question 21 of 54

Omar has been saving $\$6$ each week. Today he spent $\$90$ of the savings, and he now has $\$48$ left. For how many weeks has he been saving?

(a) Write an equation that could be used to answer the question above. First, choose the appropriate form. Then, fill in the blanks with the numbers 6, 90, and 48. Let w represent the number of weeks.

○ ___w + ___ = ___

○ ___w − ___ = ___

(b) Solve the equation in part (a) to find the number of weeks.

$w =$ _____

Question 22 of 54

Tony and Linda each opened a savings account today. Tony opened his account with a starting amount of $\$920$, and he is going to take out $\$60$ per month. Linda opened her account with a starting amount of $\$980$, and she is going to take out $\$75$ per month.

Let x be the number of months after today.

(a) For each account, write an expression for the amount of money in the account after x months. Amount of money in Tony's account (in dollars) = _____ Amount of money in Linda's account (in dollars) = _____
(b) Write an equation to show when the two accounts would have the same amount of money. _____

Question 23 of 54

Hans's Coffee Shop makes a blend that is a mixture of two types of coffee. Type A coffee costs Hans $\$5.55$ per pound, and type B coffee costs $\$4.35$ per pound. This month, Hans made 183 pounds of the blend, for a total cost of $\$899.25$. How many pounds of type B coffee did he use?

Question 24 of 54

The length of a rectangle is 5 yd longer than its width.
If the perimeter of the rectangle is 58 yd, find its length and width.

Question 25 of 54

The length of a rectangle is four times its width.
If the perimeter of the rectangle is 60 m, find its area.

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Question 26 of 54

Solve for E.

$$h = \frac{E}{mg}$$

Question 27 of 54

Solve for r.

$$5r - 9s = K$$

$r =$ ☐

Question 28 of 54

Solve for y.

$$C = (y - k)h$$

$y =$ ☐

Question 29 of 54

Solve for b.

$$L = \frac{a - b}{3}$$

$b =$ ☐

Question 30 of 54

Solve the following proportion for x.

$$\frac{8}{5} = \frac{x}{11}$$

Round your answer to the nearest tenth.

$x =$ _____

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Question 31 of 54

Solve for v.

$$\frac{6}{v-4} = \frac{8}{3}$$

Simplify your answer as much as possible.

$v =$

Question 32 of 54

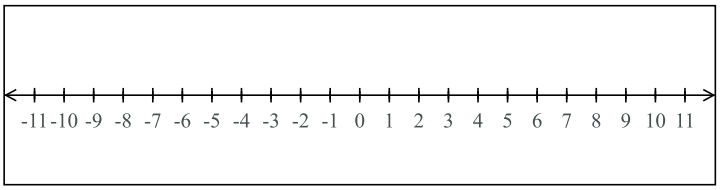
Solve for x.

$$\frac{-30}{x} = \frac{-40}{x+2}$$

Question 33 of 54

Graph the inequality below on the number line.

$$y < -2$$



Question 34 of 54

Solve the inequality for u.

$$10 \geq \frac{u}{-3}$$

Simplify your answer as much as possible.

Question 35 of 54

Solve the inequality for y.

$$3 \geq \frac{y}{4} - 12$$

Simplify your answer as much as possible.

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Question 36 of 54

Solve the inequality for y.

$$4y + 2 \geq 30 - 3y$$

Simplify your answer as much as possible.

Question 37 of 54

Solve the inequality for v.

$$-15 + 7(6 - v) > -2(v + 4)$$

Simplify your answer as much as possible.

Question 38 of 54

Abdul wants to earn at least $\$63$ trimming trees. He charges $\$6$ per hour and pays $\$9$ in equipment fees. What are the possible numbers of hours Abdul could trim trees?

Use t for the number of hours.
Write your answer as an inequality solved for t.

Question 39 of 54

To rent a certain meeting room, a college charges a reservation fee of $\$31$ and an additional fee of $\$5.90$ per hour. The history club wants to spend at most $\$72.30$ on renting the meeting room.

What are the possible amounts of time for which they could rent the meeting room?
Use t for the number of hours the meeting room is rented, and solve your inequality for t.

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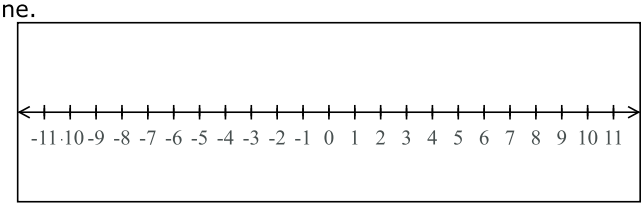
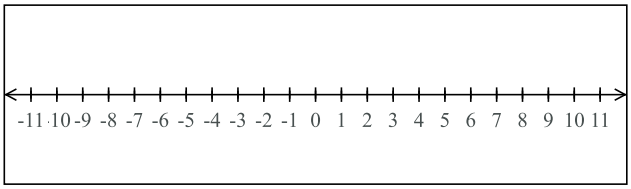
Question 40 of 54

Solve the compound inequality.

$2x - 1 > -5$ or $3x - 2 \leq -20$

Graph the solution on the number line.





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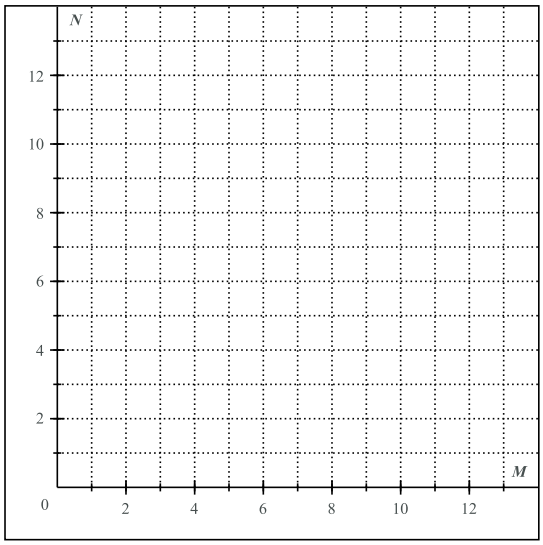
Question 41 of 54

Ali needs to read 5 novels each month.

Let N be the number of novels Ali needs to read in M months.

Write an equation relating N to M. Then graph your equation using the axes below.

Equation: _____



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Question 42 of 54

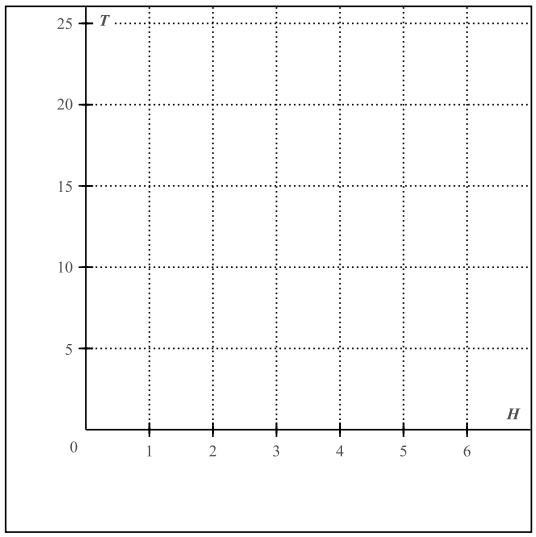
Scientists are studying the temperature on a distant planet. They find that the surface temperature at one location is 20° Celsius. They also find that the temperature *decreases* by 3° Celsius for each kilometer you go up from the surface.

Let T represent the temperature (in Celsius), and let H be the height above the surface (in kilometers).

(a) Write an equation relating T to H.

Equation:

(b) Graph your equation using the axes below.



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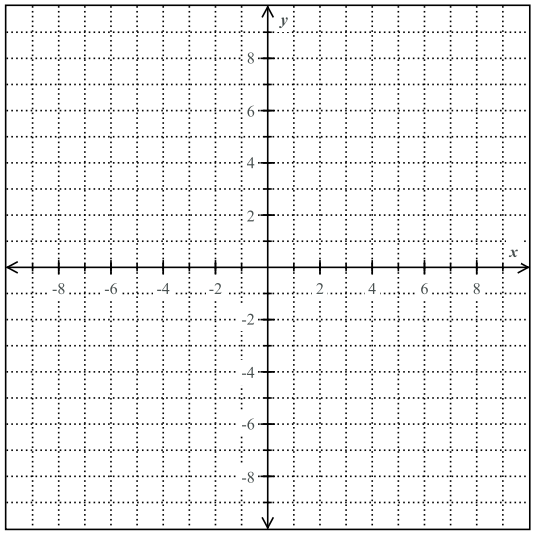
Question 43 of 54

Here is an equation.

$$-3 = 3x$$

Draw a graph to help solve the equation. Then find the solution to the equation.



Question 44 of 54

Solve for x.

$$|x| = -9$$

Question 45 of 54

Solve for x.

$$|x| + 9 = 17$$

Question 46 of 54

Solve for u.

$$|5u + 10| = 0$$

Question 47 of 54

Solve for w.

$$|w + 2| - 25 = -10$$

$w =$ ☐

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Question 48 of 54

Solve for v.

$$|3v-3|+11=32$$

Question 49 of 54

Solve for u.

$$|5u-3|=|5u-7|$$

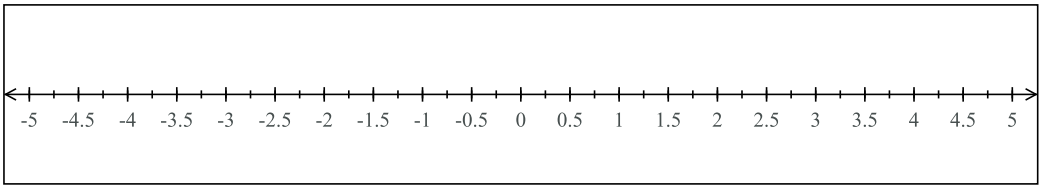
Question 50 of 54

A machine fills small bottles of glitter. The machine is set to fill each bottle with 3.75 oz of glitter, but it is acceptable to fill each bottle with 1 oz more or less than that.

(a) Let x be the amount of glitter (in ounces) in a bottle. Write an absolute value equation that can be used to find the minimum and maximum acceptable amount of glitter in a bottle. First choose an appropriate form. Then fill in the blanks.

- ○ $|__ - __| = __$
- ○ $|__ \cdot __| = __$
- ○ $|__ \div __| = __$
- ○ $|__ + __| = __$

(b) Solve the equation from part (a) and graph the solution on the number line below.



(c) Fill in the blanks.

The minimum acceptable amount of glitter in a bottle is _____ oz.

The maximum acceptable amount of glitter in a bottle is _____ oz.

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Question 51 of 54

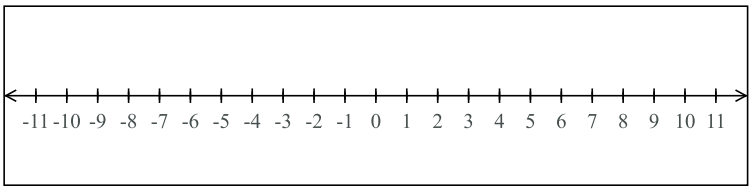
Solve.

$$3\left|y\right| \geq 33$$

Question 52 of 54

Graph the solution to the inequality on the number line.

$$\left|y-2\right| \geq 8$$



Question 53 of 54

Solve.

$$5\left|w-9\right| + 4 \geq 14$$

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Question 54 of 54

A nursery sells young palm trees. They guarantee that a palm tree will grow to be 5 feet tall, plus or minus 0.5 feet. If the tree's height doesn't meet this guarantee after 2 years, a refund can be issued.

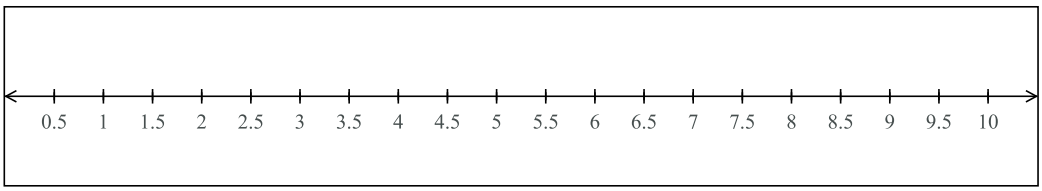
(a) Let x represent the height (in feet) of a palm tree. Write an absolute value inequality that can be used to find the *acceptable* heights of a palm tree. These are the heights where the guarantee is met.

- ○ $|___ - ___| > ___$
- ○ $|___ + ___| > ___$
- ○ $|___ + ___| \leq ___$
- ○ $|___ - ___| \leq ___$

(b) Solve the absolute value inequality from part (a).

_____

(c) Graph the solution from part (b) on the number line below.



(d) Fill in the blanks.

The palm tree will meet the guarantee if it grows to be _____ (Blank 1) feet tall, and _____ (Blank 2) feet tall.

Blank 1 Options	Blank 2 Options
less than	less than
less than or equal to	less than or equal to
greater than	greater than
greater than or equal to	greater than or equal to

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **54**

Question 1 of 54

w	Is it a solution?	
	Yes	**No**
-7	○	◉
10	○	◉
5	◉	○
-2	○	◉

Question 2 of 54

$y = -3$

Question 3 of 54

$u = 8$

Question 4 of 54

$x = 8$

Question 5 of 54

(a) Write an equation that could be used to answer the question above. First, choose the appropriate form. Then, fill in the blanks with the numbers 71, 5, and 4. Let b represent the number of books.

○ ___b + ___ = ___

◉ $4b - 5 = 71$

(b) Solve the equation in part (a) to find the number of books.

$b = 19$

Question 6 of 54

$x = \frac{W}{5y}$

Question 7 of 54

$n = \frac{-4 - 8m}{10}$

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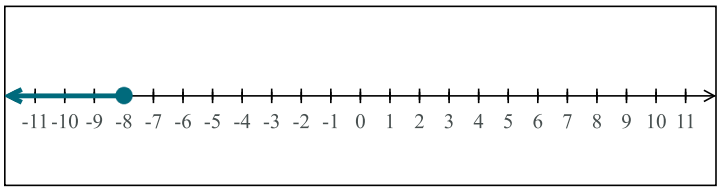
Question 8 of 54

$y = 4.1$

Question 9 of 54

$w = \frac{27}{4}$

Question 10 of 54



Question 11 of 54

$v > 10$

Question 12 of 54

$16 > y$

Question 13 of 54

y-intercept: $-\frac{3}{4}$

slope: 2

Question 14 of 54

$77\ ^\circ\text{F} = 25\ ^\circ\text{C}$

Question 15 of 54

u	Is it a solution?	
	Yes	**No**
-7	○	◉
9	○	◉
-5	○	◉
10	○	◉

Question 16 of 54

$y = -6$

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Question 17 of 54

$u = 31.25$

Question 18 of 54

Step	Reason
$\frac{y}{3} - 12 = -7$	Given equation
$\frac{y}{3} - 12 + 12 = -7 + 12$	Addition Property of Equality
$\frac{y}{3} = 5$	Simplifying
$3 \cdot \frac{y}{3} = 5 \cdot 3$	Multiplication Property of Equality
$y = 15$	Simplifying

Question 19 of 54

$y = -9$

Question 20 of 54

$x = -2$

Question 21 of 54

(a) Write an equation that could be used to answer the question above. First, choose the appropriate form. Then, fill in the blanks with the numbers 6, 90, and 48. Let w represent the number of weeks.

○ ___w + ___ = ___

◉ $6w - 90 = 48$

(b) Solve the equation in part (a) to find the number of weeks.

$w = 23$

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Question 22 of 54

(a) Writing an expression for the amount of money in each account after x months

Tony started with $\$920$ in his account and will take out $\$60$ per month.
Linda started with $\$980$ in her account and will take out $\$75$ per month.

Letting x be the number of months after today, we have the following.

Amount of money in Tony's account (in dollars) $= 920 - 60x$
Amount of money in Linda's account (in dollars) $= 980 - 75x$

(b) Writing an equation to show when the two accounts would have the same amount of money

For the two accounts to have the same amount of money, the expressions above must be equal.

So we get the following.

$$920 - 60x = 980 - 75x$$

Question 23 of 54

97 pounds

Question 24 of 54

length: 17 yd
width: 12 yd

Question 25 of 54

144m^2

Question 26 of 54

$E = mgh$

Question 27 of 54

$$r = \frac{K + 9s}{5}$$

Question 28 of 54

$$y = \frac{C}{h} + k$$

Question 29 of 54

$b = -3L + a$
We can also write this answer as $b = a - 3L$.

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Question 30 of 54

$x = 17.6$

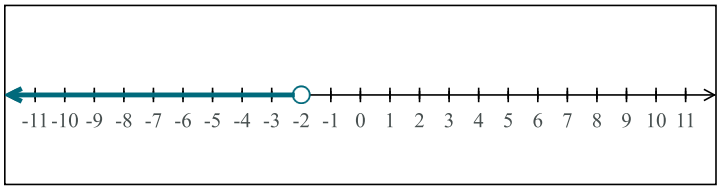
Question 31 of 54

$v = \frac{25}{4}$

Question 32 of 54

$x = 6$

Question 33 of 54



Question 34 of 54

$-30 \leq u$

Question 35 of 54

$60 \geq y$

Question 36 of 54

$y \geq 4$

Question 37 of 54

$v < 7$

The answer can also be written as $7 > v$.

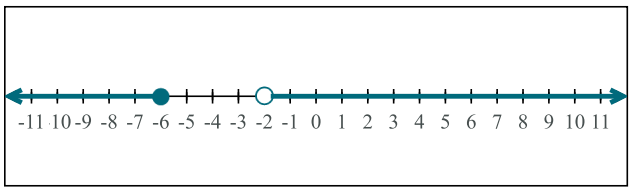
Question 38 of 54

$t \geq 12$

Question 39 of 54

$t \leq 7$

Question 40 of 54



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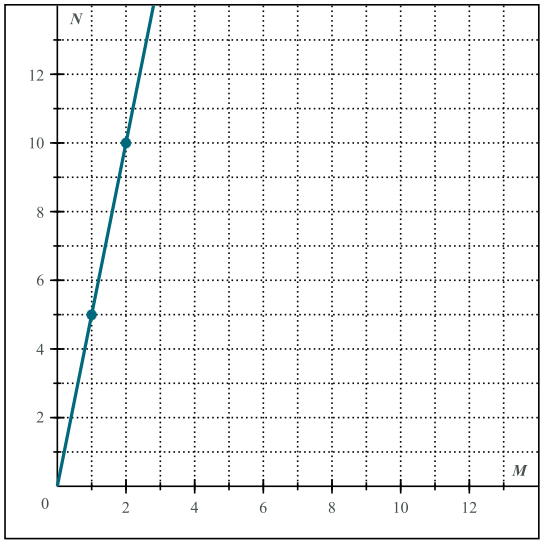
Question 41 of 54

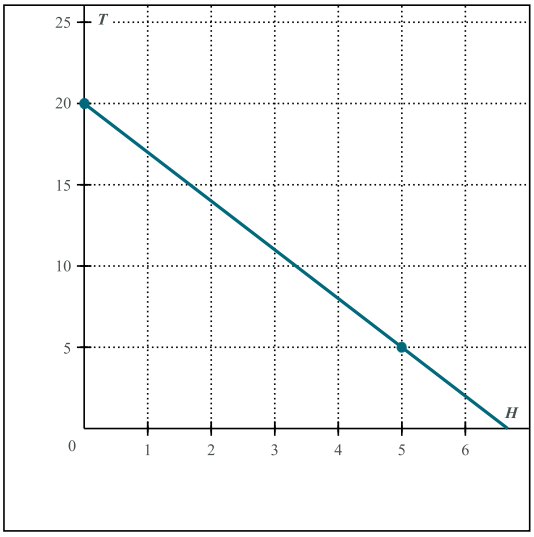
Equation: $N = 5M$



Question 42 of 54

(a) Equation: $T = 20 - 3H$

(b)



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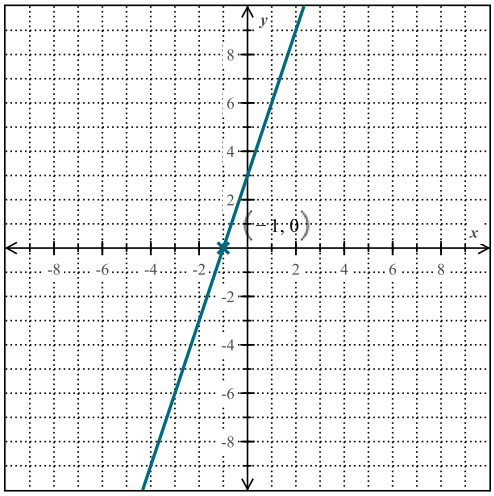
Question 43 of 54



Solution: $x = -1$

Question 44 of 54

No solution

Question 45 of 54

$x = 8,\ -8$

Question 46 of 54

$u = -2$

Question 47 of 54

$w = 13,\ -17$

Question 48 of 54

$v = 8,\ -6$

Question 49 of 54

$u = 1$

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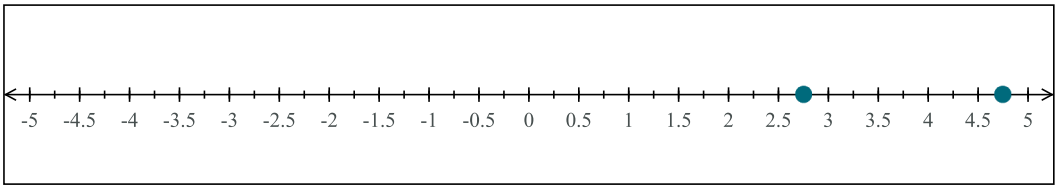
Question 50 of 54

(a)

- (●) $|x - 3.75| = 1$
- ( ) $|\square \cdot \square| = \square$
- ( ) $|\square \div \square| = \square$
- ( ) $|\square + \square| = \square$

(b)

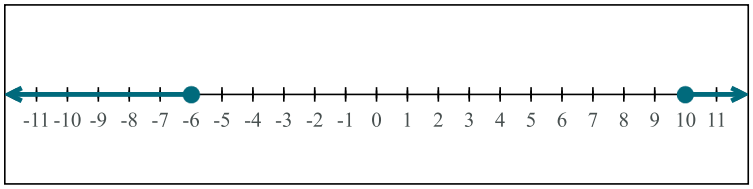


(c) The minimum acceptable amount of glitter in a bottle is 2.75 oz.
The maximum acceptable amount of glitter in a bottle is 4.75 oz.

Question 51 of 54

$y \leq -11$ or $y \geq 11$

Question 52 of 54



Question 53 of 54

$w \leq 7$ or $w \geq 11$

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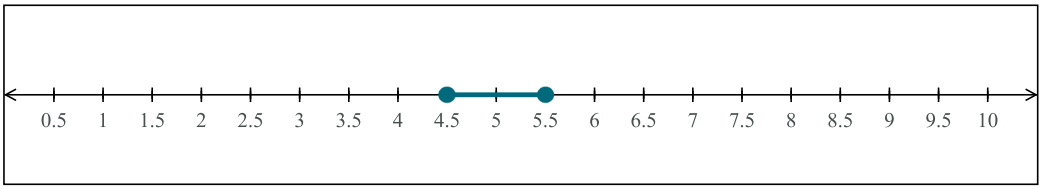
Question 54 of 54

(a)

- ○ $|\square - \square| > \square$
- ○ $|\square + \square| > \square$
- ○ $|\square + \square| \leq \square$
- ◉ $|x - 5| \leq 0.5$

(b) $4.5 \leq x \leq 5.5$

(c)



(d) The palm tree will meet the guarantee if it grows to be [greater than or equal to] 4.5 feet tall, and [less than or equal to] 5.5 feet tall.

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